

Newton Iteration.

Prop. Let $\alpha > 0$ be fixed. Given $x_1 > \sqrt{\alpha}$, define recursively

$$x_{n+1} = \frac{1}{2} \cdot \left(x_n + \frac{\alpha}{x_n} \right) \quad (n \in \mathbb{N})$$

Then, $\{x_n\}$ is monotonically decreasing and converges to $\sqrt{\alpha}$.

Proof. Observe that: for each $n \in \mathbb{N}$,

$$\frac{1}{2} \left(x_n + \frac{\alpha}{x_n} \right) = x_{n+1} \leq x_n \iff \alpha \leq x_n^2 \quad \text{if } x_n > 0.$$

(Clearly, all $x_n > 0$, by definition, so using $\alpha \leq x_n^2$ for all $n \in \mathbb{N}$, by induction, For $n=1$, $\alpha \leq x_1^2$, since $\sqrt{\alpha} < x_1$. So, $x_2 \leq x_1$.

If $\alpha \leq x_n^2$, then

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{\alpha}{x_n} \right) \leq \frac{1}{2} \cdot (x_n + x_n) = x_n$$

and,

$$\frac{1}{2} \left(x_n + \frac{\alpha}{x_n} \right) \geq \sqrt{x_n \cdot \frac{\alpha}{x_n}} = \sqrt{\alpha}, \quad \text{so } \sqrt{\alpha} \leq x_{n+1} \leq x_n.$$

So that, the equivalent condition with the induction gives that

$$x_{n+1} \leq x_n \quad \text{for all } n \in \mathbb{N}$$

That is, decrease monotonically, and $x_n \geq \sqrt{\alpha}$ for all $n \in \mathbb{N}$

Now, since $\{x_n\}$ is bounded below by $\sqrt{\alpha}$ and decreasing,

$\{x_n\}$ has a limit, denote $L = \lim_{n \rightarrow \infty} x_n$. Then,

$$L = \lim_{n \rightarrow \infty} x_{n+1} = \lim_{n \rightarrow \infty} \frac{1}{2} \left(x_n + \frac{\alpha}{x_n} \right) = \frac{1}{2} L + \frac{\alpha}{2L} \Rightarrow L = \sqrt{\alpha}.$$

So, proved $\square$

Note: Moreover, we can measure the 'error': put $\epsilon_n = |x_n - \sqrt{\alpha}| = x_n - \sqrt{\alpha}$.

Then, for each $n \in \mathbb{N}$, $\epsilon_{n+1} = \frac{\epsilon_n^2}{2x_n} < \frac{\epsilon_n^2}{2\sqrt{\alpha}}$ and that $\epsilon_{n+1} < 2\sqrt{\alpha} \cdot \left(\frac{\epsilon_1}{2\sqrt{\alpha}} \right)^{2^n}$.

Since

$$\begin{aligned} \epsilon_{n+1} &= x_{n+1} - \sqrt{\alpha} = \frac{1}{2} \cdot \left(x_n + \frac{\alpha}{x_n} \right) - \sqrt{\alpha} = \frac{1}{2} \cdot \left(\epsilon_n + \sqrt{\alpha} + \frac{\alpha}{\epsilon_n + \sqrt{\alpha}} \right) - \sqrt{\alpha} \\ &= \frac{1}{2} \cdot \left(\frac{\epsilon_n^2}{\epsilon_n + \sqrt{\alpha}} \right) = \frac{\epsilon_n^2}{2x_n} < \frac{\epsilon_n^2}{2\sqrt{\alpha}}, \quad \text{so that} \end{aligned}$$

$$\epsilon_n < \frac{1}{2\sqrt{\alpha}} \cdot \epsilon_{n-1}^2 < \frac{1}{2\sqrt{\alpha}} \cdot \left(\frac{1}{2\sqrt{\alpha}} \right)^2 \cdot \epsilon_{n-2}^2 \leq \dots \leq \left(\frac{1}{2\sqrt{\alpha}} \right)^{2^n - 1} \cdot \epsilon_1^{2^n} \quad \square$$

Thm. Let $g, f_n: (0, \infty) \rightarrow \mathbb{R}$ be Riemann-integrable on $[t, T]$ for any $0 < t < T$.
and satisfies that $0 \leq |f_n| \leq g$ for all $n \in \mathbb{N}$, and
 $f_n \rightarrow f$ uniformly on every compact subset of $(0, \infty)$, and
$$\int_0^\infty g(x) dx < \infty.$$

Then,
$$\lim_{n \rightarrow \infty} \int_0^\infty f_n(x) dx = \int_0^\infty f(x) dx.$$

Proof.

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Thm. Let $\{f_n: \mathbb{R} \rightarrow \mathbb{R}\}$ be a sequence of continuous functions.

If $f_n \rightarrow f$ pointwise for all $x \in \mathbb{R}$, then the set

$\{x \in \mathbb{R} \mid f \text{ is continuous at } x\}$

is uncountable.

Proof.

Thm. Let K be a Compact Metric Space.

Suppose that $\{f_n: K \rightarrow \mathbb{R}\}$ is a sequence of Continuous function, which Converges Continuous function f pointwise on K and satisfying: for each $n \in \mathbb{N}$,

$$f_n(x) \geq f_{n+1}(x) \text{ for all } x \in K.$$

Then, $f_n \rightarrow f$ uniformly on K .

Proof. Put $g_n := f_n - f$. Then, g_n is continuous and $g_n \rightarrow 0$ pointwise, and $g_n = f_n - f \geq f_{n+1} - f = g_{n+1}$.

Let $\varepsilon > 0$ be given. Then, the set

$$K_n := \{x \in K \mid g_n(x) \geq \varepsilon\} = g_n^{-1}([\varepsilon, \infty))$$

is closed, because g_n is continuous function.

And, since $g_n(x) \geq g_{n+1}(x)$ for all $x \in K$, $g_{n+1}(x) \geq \varepsilon \Rightarrow g_n(x) \geq \varepsilon$, so

$$K_{n+1} \subseteq K_n, \text{ for each } n \in \mathbb{N}.$$

Let $x \in \bigcap_{n \in \mathbb{N}} K_n$ be fixed. Since $g_n(x) \rightarrow 0$ as $n \rightarrow \infty$, for some $N \in \mathbb{N}$, $x \notin K_N$ and therefore $K = \emptyset$. But, since each K_n is closed set in the Compact set K , so K_n is Compact set, thus

$$\bigcap_{n \in \mathbb{N}} K_n = \emptyset \Rightarrow \text{for some } N \in \mathbb{N}, K_N = \emptyset.$$

That is, for N ,

$$g_n(x) < \varepsilon \text{ for all } x \in K,$$

It implies $\sup_{x \in K} |g_n(x)| = \sup_{x \in K} |f_n(x) - f(x)| \rightarrow 0$ as $n \rightarrow \infty$. $\square$

Topology.

Let Y be a Hausdorff space, and let X be any space.

Assume $f, g: X \rightarrow Y$ are continuous maps and $f|_A = g|_A$ for a dense set $A \subset X$.
Prove that $f = g$.

Proof. Let $x \in X$ be given. If $f(x) = g(x)$, then we are done. Assume $f(x) \neq g(x)$.
Since Y is T_2 space, there exist disjoint open sets U, V such that
 $f(x) \in U, g(x) \in V, U \cap V = \emptyset$.

Since f, g are continuous, the preimages are also open, so that

$$x \in f^{-1}[U] \cap g^{-1}[V]$$

so, these $f^{-1}[U] \cap g^{-1}[V]$ is non-empty open set, since $A \subset X$ is dense in X ,

$$A \cap (f^{-1}[U] \cap g^{-1}[V]) \neq \emptyset,$$

Equivalently, there is $a \in A$ such that $f(a) \in U$ and $g(a) \in V$.

But, by the hypothesis, $f(a) = g(a)$, so that $U \cap V \neq \emptyset$, it contradicts.
Consequently, $f(x) = g(x)$ for all $x \in X$. $\square$

Weierstrass M-test.

Thm. Let $f_n: X \rightarrow \mathbb{R}$ ($n \in \mathbb{N}$) be functions on a space X such that for some $M_n > 0$, $|f_n(x)| \leq M_n$ for all $x \in X$.

If $\sum_{n=1}^{\infty} M_n$ converges, then $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly on X .

Proof. For any $m > n$,

$$\left| \sum_{k=1}^m f_k(x) - \sum_{k=1}^n f_k(x) \right| = \left| \sum_{k=n+1}^m f_k(x) \right| \leq \sum_{k=n+1}^m |f_k(x)| \leq \sum_{k=n+1}^m M_k$$